Machine Learning Research Engineering Portfolio

2026 Edition

Derek Lankeaux, MS Applied Statistics
Data Scientist | Applied Statistician | ML Research Engineer
Rochester Institute of Technology

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AI Standards Compliance: IEEE 2830-2025, ISO/IEC 23894:2025, EU AI Act (2025)

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Abstract

This portfolio presents three end-to-end machine learning research engineering projects demonstrating 2026-grade competencies in applied ML, Bayesian inference, GenAI evaluation, and responsible AI deployment. Project 1 develops a scalable AI safety red-team evaluation framework combining LLM ensemble annotation (Krippendorff's $\alpha = 0.81$) with supervised ML classification achieving 0.9923 ROC-AUC and a $340\times$ cost reduction over human annotation. Project 2 implements an ensemble learning pipeline for breast cancer classification achieving 99.12% accuracy and 0.9987 ROC-AUC with full calibration analysis and clinical decision framing. Project 3 applies a Bayesian hierarchical framework to detect publisher-level political bias in educational textbooks using LLM consensus scoring ($\alpha = 0.84$), yielding credible posterior estimates with 95% Highest Density Intervals (HDI). Across all three projects, consistent emphasis is placed on statistical rigor, reproducibility, explainability, and production readiness.

Keywords: Machine Learning, Applied Statistics, Bayesian Inference, Ensemble Methods, LLM Evaluation, AI Safety, Breast Cancer Classification, Bias Detection, XGBoost, PyMC, SHAP, MLOps, Responsible AI

Contents

1 Portfolio Overview

1.1 Research and Engineering Contributions

This portfolio demonstrates the ability to design, implement, evaluate, and communicate production-grade ML systems across three applied domains. Core contributions include:

- **Reproducible ML workflows** with explicit data versioning, experiment tracking, and deterministic evaluation pipelines
- **Statistically grounded evaluation**: confidence intervals, subgroup analysis, calibration diagnostics, and ablation studies
- **Engineering best practices**: modular code, test coverage, CI pipelines, and inference/runtime profiling
- **Decision-oriented metrics** that map model performance to real-world operational objectives
- **Responsible AI documentation**: limitations, governance controls, and human-in-the-loop escalation policies

1.2 Project Summary

Table 1: Summary of three portfolio projects

Project	Domain	Primary Method	Key Result
AI Safety Red-Team Evaluation	AI Governance	LLM Ensemble + ML Classification	ROC-AUC 0.9923; $\alpha =$
Breast Cancer Classification	Clinical ML	Ensemble Learning + Calibration	Accuracy 99.12%; ROC
LLM Ensemble Bias Detection	NLP / Bayesian	Hierarchical Bayes + LLM Consensus	$\alpha = 0.84$; 3/5 publishers

1.3 Shared Methodology Standards

All three projects share a common methodological foundation:

1. **Pre-registration equivalents**: defined hypotheses, metrics, and analysis plans before fitting final models
2. **Held-out test sets**: no leakage between feature engineering, model selection, and final evaluation
3. **Uncertainty quantification**: 95% confidence intervals (frequentist) or 95% HDI (Bayesian) on all estimates
4. **Explainability**: SHAP-based global and local feature attribution for tree-based models
5. **Standards compliance**: IEEE 2830-2025 transparent ML documentation, ISO/IEC 23894:2025 AI risk management

2 Project 1: AI Safety Red-Team Evaluation

2.1 Problem Statement

Manual red-teaming is expensive ($50–100/hour per human expert), non-scalable, and subject to inconsistent inter-annotator agreement (70–85%). Modern LLM release cycles demand evaluation throughput that human red-teamers cannot sustain. This project develops and validates a hybrid human-AI evaluation framework achieving annotation quality comparable to expert consensus at a 340× cost reduction.

2.2 Technical Framework

2.2.1 Stage 1 — LLM Ensemble Annotation

Three frontier LLMs serve as calibrated annotators: GPT-4o, Claude-3.5-Sonnet, and Llama-3.2-90B. A structured prompt protocol enforces a six-category harm taxonomy (Dangerous Information, Hate/Discrimination, Deception/Manipulation, Privacy Violation, Illegal Activity, Self-Harm/Violence). Inter-rater reliability is validated with Krippendorff's $\alpha = 0.81$ (Excellent threshold: ≥ 0.80). The corpus comprises 12,500 AI model response pairs processed at $\sim$180 samples/hour per LLM.

2.2.2 Stage 2 — ML Classification Pipeline

Eight ensemble classifiers are trained on LLM-generated labels: Random Forest, Gradient Boosting, AdaBoost, Bagging, XGBoost, LightGBM, Voting, and Stacking. A feature engineering pipeline produces 47 features including lexical patterns, semantic embeddings, structural cues, and LLM confidence signals. Class imbalance is mitigated via SMOTE oversampling and class-weighted loss. The best model (Stacking Classifier) processes $\sim$850 samples/hour at $0.018/sample.

2.2.3 Bayesian Hierarchical Risk Modeling

A PyMC-based hierarchical model with partial pooling operates over harm categories and AI model vendors. 95% HDI quantifies uncertainty on category-level harm rates; posterior predictive checks confirm model calibration.

2.3 Results

Table 2: LLM annotation reliability metrics

Metric	Value	Interpretation
Krippendorff's α (overall)	0.81	Excellent inter-rater agreement
GPT-4o $\leftrightarrow$ Claude-3.5 (r)	0.94	Near-perfect pairwise agreement
GPT-4o $\leftrightarrow$ Llama-3.2 (r)	0.88	Excellent pairwise agreement
Claude-3.5 $\leftrightarrow$ Llama-3.2 (r)	0.86	Excellent pairwise agreement
Fleiss' κ (3-way)	0.79	Substantial agreement

Table 3: Stage 2 classification performance (Stacking Classifier)

Metric	Value
Accuracy	96.8%
Precision	97.2%
Recall (Sensitivity)	96.1%
F1-Score	96.6%
ROC-AUC	0.9923
Cross-Validation (10-fold)	95.9% $\pm$ 1.4%

Table 4: Defense effectiveness against adversarial prompts

Defense Configuration	Harm Rate	Reduction vs. Baseline
No defense (baseline)	21.8%	—
Single-stage filter	11.4%	47.7%
Dual-stage filter	4.8%	78.0%
+ Adversarial fine-tuning	2.1%	90.4%

2.4 Production Architecture

Throughput: $\sim$850 prompt-response pairs/hour (combined pipeline). Cost: \$0.018/sample versus $\sim$\$6.25/sample human baseline. P95 latency: < 2.1 seconds end-to-end. API: REST endpoint with async queue; 99.7% uptime SLA target.

2.5 Threats to Validity

- LLM annotators share training corpus overlap, which may inflate agreement
- Harm taxonomy reflects April 2026 standards; regulatory updates may require re-categorization
- Evaluation corpus is synthetic/curated and may not represent the full production adversarial distribution
- Dual-filter defense effectiveness is measured in controlled conditions; real-world performance may differ

3 Project 2: Breast Cancer Classification

3.1 Problem Statement

Breast cancer is the most prevalent malignancy among women globally ($\sim$2.3 million new diagnoses annually). Fine Needle Aspiration (FNA) cytology interpretation exhibits inter-observer variability (85–95% concordance), creating demand for consistent, reproducible computer-aided diagnosis support. This project develops and validates an ensemble ML system for binary malignancy classification from FNA cytological features, with explicit clinical threshold analysis.

3.2 Technical Framework

3.2.1 Data Engineering

Dataset: Wisconsin Diagnostic Breast Cancer (WDBC), $n = 569$ (357 benign, 212 malignant). Features: 30 cytological measurements (10 nuclear characteristics $\times$ 3 statistical summaries). Preprocessing includes Variance Inflation Factor (VIF) multicollinearity analysis and StandardScaler normalization. SMOTE oversampling is applied to training folds only (leakage-safe). Recursive Feature Elimination (RFE) retains 17 features.

3.2.2 Ensemble Benchmarking

Eight algorithms are evaluated: Random Forest, Gradient Boosting, AdaBoost, Bagging, XGBoost, LightGBM, Voting, and Stacking. Hyperparameter optimization uses Optuna Bayesian search (200 trials, TPE sampler). Validation: 10-fold stratified cross-validation plus a held-out test set ($n = 114$, 20%).

3.2.3 Calibration Analysis

Pre- and post-calibration metrics include Expected Calibration Error (ECE), Brier score, and reliability curves. Calibration methods compared: Platt scaling (sigmoid) and isotonic regression. Clinical threshold analysis evaluates the sensitivity–specificity trade-off at 50 operating points.

3.3 Results

Table 5: Classification performance on held-out test set (AdaBoost)

Metric	**Value**	**Clinical Interpretation**
Accuracy	99.12%	113/114 correct classifications
Precision (PPV)	100.00%	Zero false positives
Recall (Sensitivity)	98.59%	1 missed malignancy in 71 cases
Specificity	100.00%	Perfect benign identification
F1-Score	99.29%	Harmonic mean balance
ROC-AUC	0.9987	Near-perfect discrimination
Cohen's κ	0.9823	Almost perfect agreement

95% CI (test accuracy): [96.27%, 100.65%]; Binomial test vs. random baseline: $p < 0.0001$.

Table 6: Calibration improvement after Platt scaling (AdaBoost)

Metric	**Before Calibration**	**After Platt Scaling**
Expected Calibration Error (ECE)	0.0312	0.0089
Brier Score	0.0421	0.0187
Max Calibration Error	0.0847	0.0234

Table 7: Clinical threshold analysis: key operating points

Operating Point	**Sensitivity**	**Specificity**	**PPV**	**NPV**
Default (0.50)	98.59%	100.00%	100.00%	97.67%
High Sensitivity (0.30)	100.00%	97.62%	98.61%	100.00%
High Specificity (0.70)	95.77%	100.00%	100.00%	95.45%

Recommended operating point for screening: threshold = 0.30 (zero missed malignancies).

3.4 Top Predictive Features (SHAP)

Table 8: Top 5 features by mean absolute SHAP value

Rank	**Feature**	**SHAP Mean Abs**	**Clinical Relevance**
1	Worst Radius	0.847	Tumor size indicator
2	Worst Concave Points	0.723	Nuclear irregularity
3	Mean Concave Points	0.641	Shape complexity
4	Worst Perimeter	0.589	Boundary length
5	Mean Radius	0.521	Average cell size

3.5 Threats to Validity

- WDBC dataset (1995) may not reflect contemporary cytology imaging pipelines
- Single-institution data limits generalizability across demographic and equipment variation
- 10-fold CV may be optimistic on small dataset ($n = 569$); external validation not performed
- SMOTE introduces synthetic samples; real-world class distribution may differ from training split

4 Project 3: LLM Ensemble Bias Detection

4.1 Problem Statement

Political bias in educational textbooks has long-term effects on civic socialization. Traditional audit methods are subjective, expensive, and non-scalable. This project develops a scalable, reproducible computational audit framework using LLM consensus scoring and Bayesian hierarchical modeling to detect and quantify publisher-level political framing in K–12 educational content.

4.2 Technical Framework

4.2.1 Corpus Construction

150 textbooks from 5 major educational publishers (Publishers A–E). 4,500 passages sampled (30 per textbook) across four topics: government, economics, history, and social issues. 67,500 bias ratings generated (4,500 passages $\times$ 3 LLMs $\times$ 5 dimensions).

4.2.2 LLM Ensemble Protocol

Three frontier LLMs: GPT-4, Claude-3-Opus, Llama-3-70B. Structured prompt: passage + rating rubric (-2 strongly liberal to $+2$ strongly conservative). Deterministic decoding (temperature = 0) for reproducibility across five dimensions: word choice, framing, source selection, omission patterns, and tone.

4.2.3 Bayesian Hierarchical Model

Three-level hierarchy: dimension $\rightarrow$ publisher $\rightarrow$ textbook $\rightarrow$ passage. Partial pooling provides shrinkage toward grand mean, preventing overfitting to small publishers. MCMC inference: PyMC, 4 chains $\times$ 2,000 draws (1,000 warm-up), NUTS sampler. Convergence: R-hat < 1.01 on all parameters; ESS $> 3,000$.

4.3 Results

Table 9: Inter-rater reliability across three LLMs

Metric	Value	Interpretation
Krippendorff's α (3-LLM)	0.84	Excellent (≥ 0.80 threshold)
GPT-4 $\leftrightarrow$ Claude-3 (r)	0.92	Near-perfect
GPT-4 $\leftrightarrow$ Llama-3 (r)	0.89	Excellent
Claude-3 $\leftrightarrow$ Llama-3 (r)	0.87	Excellent
Fleiss' κ (3-way)	0.81	Substantial to excellent

Table 10: Publisher bias estimates from Bayesian hierarchical model

Rank	Publisher	Posterior Mean	95% HDI	Classification
1	Publisher C	-0.48	$[-0.62, -0.34]$	**Credibly Liberal**
2	Publisher A	-0.29	$[-0.41, -0.17]$	**Credibly Liberal**
3	Publisher E	$+0.02$	$[-0.10, +0.14]$	Neutral
4	Publisher B	$+0.08$	$[-0.04, +0.20]$	Neutral
5	Publisher D	$+0.38$	$[+0.26, +0.50]$	**Credibly Conservative**

Classification rule: credible effect when 95% HDI excludes zero. Scale: -2 (strongly liberal) to $+2$ (strongly conservative).

Table 11: MCMC convergence diagnostics

Parameter Group	R-hat	ESS (bulk)	ESS (tail)
Publisher effects	1.001	4,218	3,891
Textbook effects	1.002	3,744	3,512
Dimension effects	1.001	4,106	3,988
Grand mean	1.000	5,201	4,873

4.4 Threats to Validity

- LLMs may share training data with evaluated textbooks, introducing systematic rating bias
- Five-publisher corpus may not generalize to independent, state-specific, or digital-native publishers
- Political bias scale is ordinal; interval assumptions in Bayesian model are approximations
- LLM political calibration is time-sensitive; model updates may alter future ratings without replication

5 Cross-Project Technical Strengths

5.1 Statistical Rigor

Table 12: Statistical capabilities by project

Capability	Project 1	Project 2	Project 3
Confidence/HDI intervals	✓(95% HDI)	✓(95% CI)	✓(95% HDI)
Cross-validation	✓(10-fold)	✓(10-fold)	N/A
Inter-rater reliability	✓($\alpha = 0.81$)	N/A	✓($\alpha = 0.84$)
Calibration analysis	✓	✓(ECE, Brier)	N/A
Bayesian inference	✓(hierarchical)	N/A	✓(hierarchical)
Multiple testing adjustment	✓	N/A	✓(Dunn/BH)

5.2 Responsible AI Controls

All three projects include:

- **Limitations disclosure**: explicit threats to validity and generalizability caveats
- **Human-in-the-loop**: configurable escalation thresholds and audit logging

- **Governance documentation**: model cards, intended-use statements, misuse risk assessment
- **Uncertainty communication**: all estimates carry quantified uncertainty intervals

6 Role Alignment and Competency Map

Table 13: Competency evidence map

Competency	Evidence
Supervised ML (classification)	Projects 1 & 2: 8+ ensemble algorithms, full benchmark
Bayesian inference	Projects 1 & 3: PyMC hierarchical models with MCMC
LLM evaluation & prompting	Projects 1 & 3: multi-LLM annotation, structured prompts
Calibration & uncertainty	Project 2: ECE, Brier, Platt scaling
Statistical hypothesis testing	Projects 2 & 3: binomial, Kruskal–Wallis, Dunn, Friedman
Explainability (SHAP)	Projects 1 & 2: global + local attribution
MLOps & deployment	Projects 1 & 2: REST API, latency profiling, drift monitoring
Responsible AI	All three: governance docs, limitations, escalation policies

Table 14: Applicable role alignment

Role	Alignment
Data Scientist (Applied ML)	Primary (Projects 1 & 2)
Applied Statistician	Primary (Projects 2 & 3)
GenAI Evaluation Scientist	Primary (Projects 1 & 3)
ML Research Engineer	All three projects
AI Safety / Red Team Researcher	Project 1

7 Reproducibility and Deployment Notes

7.1 Code and Data Availability

Table 15: Repository resources

Resource	Location
Repository	`github.com/dl1413/Machine-Learning-Research-Engineering-Proj`
AI Safety Notebook	`AI_Safety_RedTeam_Evaluation.ipynb`
Breast Cancer Notebook	`Breast_Cancer_Classification_PUBLICATION.ipynb`
Bias Detection Notebook	`LLM_Ensemble_Textbook_Bias_Detection.ipynb`
PDF Build Pipeline	`scripts/build_reports_pdf.sh`

7.2 Reproducibility Controls

All projects share: fixed random seeds (seed = 42 throughout), pinned dependency versions (requirements files), deterministic preprocessing pipelines, and environment specification (Python 3.11, conda/pip lockfiles).

7.3 PDF Generation

Publication-ready PDFs are generated via:

```
chmod +x scripts/build_reports_pdf.sh
./scripts/build_reports_pdf.sh
```

Output PDFs are in `pdf/out/`. See `PDF_EXPORT.md` for full build instructions.

8 Conclusions

This portfolio demonstrates end-to-end machine learning research engineering capability across three distinct applied domains:

1. **AI Safety Red-Team Evaluation** establishes a cost-effective, audit-grade framework for scalable LLM harm detection, reducing annotation cost by 340$\times$ while preserving expert-level reliability ($\alpha = 0.81$, ROC-AUC 0.9923).
2. **Breast Cancer Classification** delivers clinically viable diagnostic support with exceptional discriminative performance (ROC-AUC 0.9987) and improved probability calibration (ECE reduced from 0.0312 to 0.0089), enabling threshold-optimized operating points for screening versus diagnosis contexts.
3. **LLM Ensemble Bias Detection** provides a reproducible, statistically rigorous audit methodology for educational content, identifying credible political framing effects in 3 of 5 publishers with an uncertainty-aware triage protocol for expert review.

Together, these projects reflect competencies in ML systems design, statistical inference, responsible AI governance, and engineering rigor required for senior data science and applied ML research roles in 2026.

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